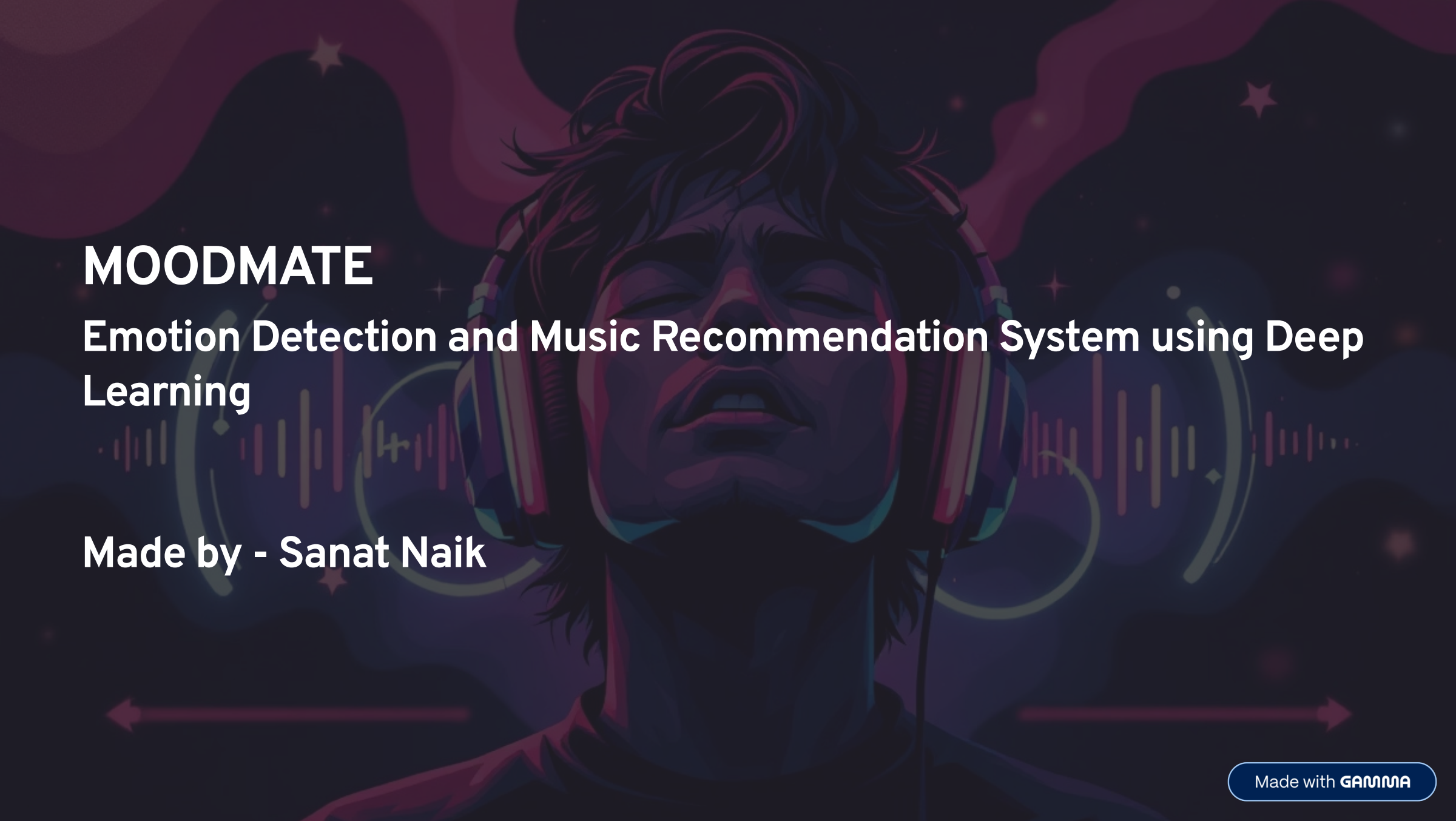




MOODMATE

Emotion Detection and Music Recommendation System using Deep Learning

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PROJECT OVERVIEW

OBJECTIVE: To develop an intelligent system that detects a user's emotional state from text input and recommends music that aligns with the detected emotion.

Deep Learning Model

A deep learning model capable of classifying emotions from text

Recommendation Engine

A content-based music recommendation engine

Interactive Interface

An interactive interface providing real-time emotion-based music suggestions

OUTCOMES:

- A deep learning model capable of classifying emotions from text
- A content-based music recommendation engine
- An interactive interface providing real-time emotion-based music suggestions

DATASET OVERVIEW AND KEY INSIGHTS

Emotion Dataset:

- Structure: Sentence, Emotion label
- Size: ~130,000+ text samples
- Classes: Reduced to 5 core emotions (happy, sad, calm, energetic, angry)

FAST.FM Music Dataset:

- Metadata: Artist, song name, Spotify ID, tags
- Tags used to represent genre and mood information



REASON FOR CHOOSING DATASETS:

- Rich textual data suitable for emotion classification
- Music tags allow semantic mapping between emotions and songs
- Well-suited for supervised learning and recommendation systems

METHODOLOGY



DATA PREPROCESSING

Cleaning and preparing raw text data



FEATURE EXTRACTION

Converting text into numerical representations



MODEL ARCHITECTURE

Designing deep learning models for emotion detection



TRAINING AND EVALUATION

Evaluating performance using standard metrics

DATA PREPROCESSING & FEATURE EXTRACTION

DATA PREPROCESSING

1. TEXT CLEANING:

- Removed sentences without labels
- Removed special characters and extra spaces
- Converted text to lowercase for normalization

2. LABEL SIMPLIFICATION:

- Original emotion labels were reduced to 5 core emotions
- Simplification improves model focus and music mapping

FEATURE EXTRACTION

1. TOKENIZATION:

- Text converted into tokens using a pretrained tokenizer
- Enables numerical processing by the neural network

2. PADDING:

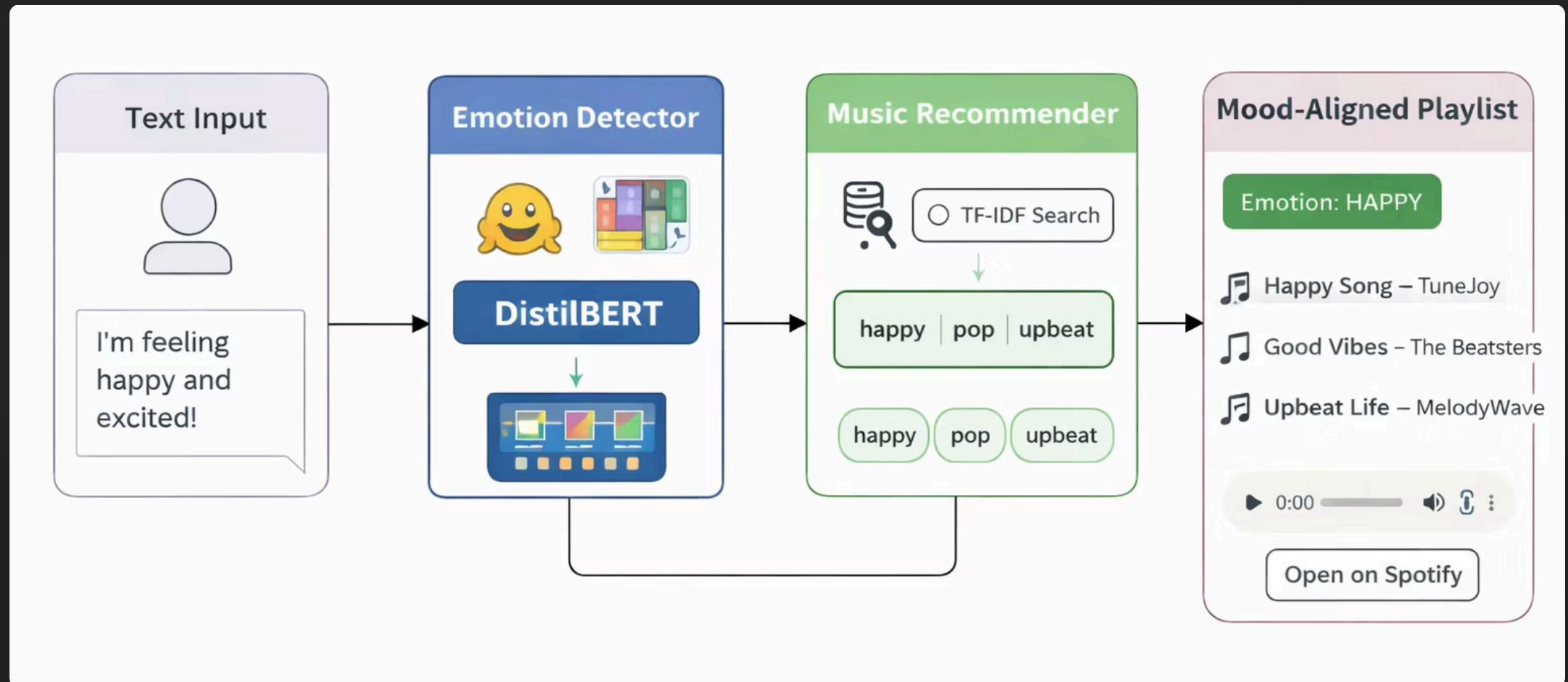
- All sequences padded or truncated to a fixed length
- Ensures uniform input size for batch training

3. LABEL ENCODING:

- Emotion labels converted to numeric format
- Enables multi-class classification

MODEL ARCHITECTURE

The system uses a **Transformer-based architecture (DistilBERT)** with a **custom deep classification head**.



LAYERS:

- Pretrained DistilBERT encoder
- Fully connected dense layers
- ReLU activation and dropout for regularization
- Output layer for emotion classification

WHY DISTILBERT:

- Lightweight and faster than BERT
- Maintains strong performance
- Suitable for real-time applications

EXPLORATORY DATA ANALYSIS & VISUALIZATION

EXPLORATORY DATA ANALYSIS

- Identified class imbalance in emotion distribution
- Positive emotions appeared more frequently than negative ones
- Dataset size sufficient for robust training

VISUALIZATION

- Emotion distribution plots revealed imbalance
- Training vs validation loss curves used to analyze overfitting
- Accuracy and F1-score trends monitored across epochs

TRAINING & EVALUATION

Dataset split:

- 70% Training
- 15% Testing
- 15% Validation

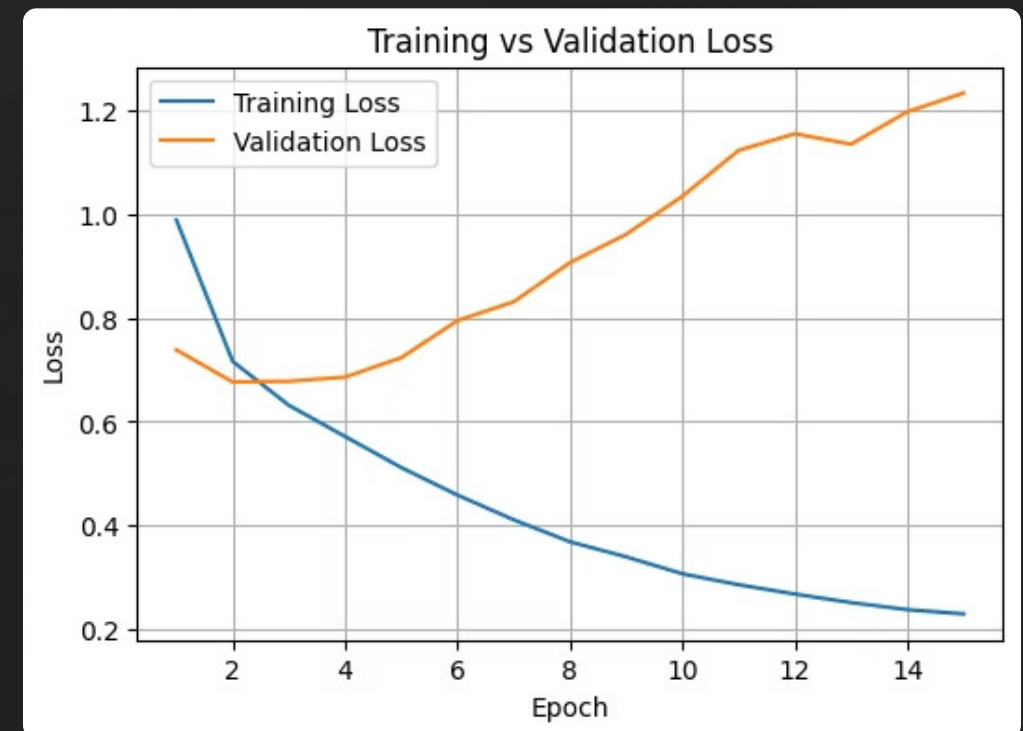
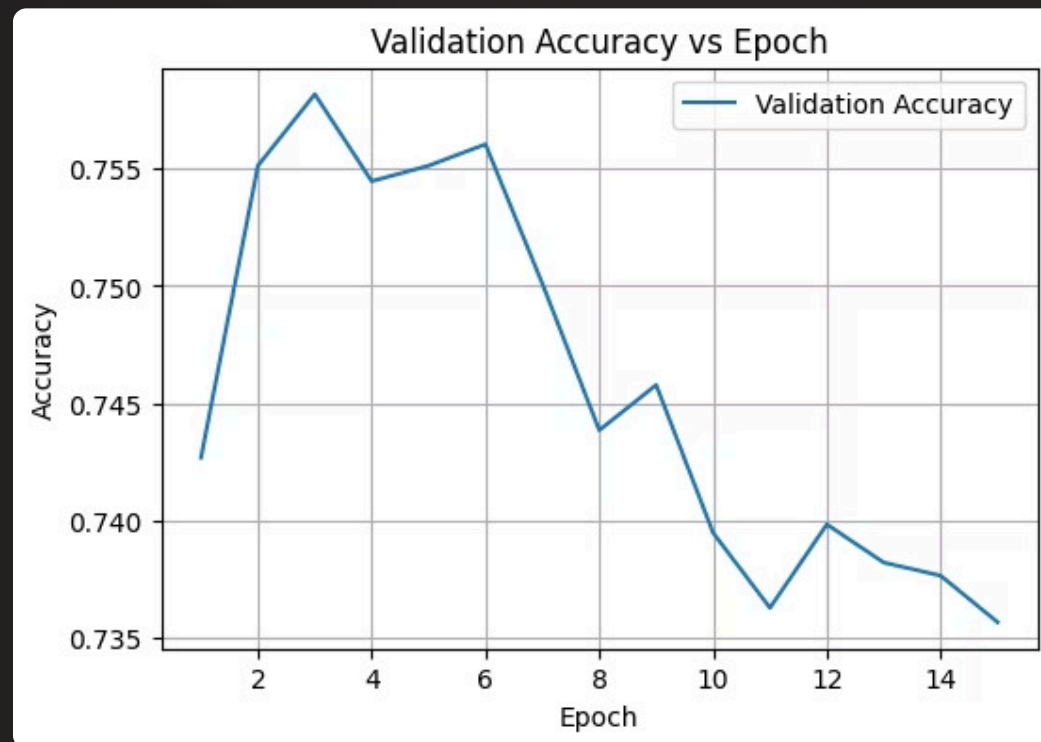
Validation performed during training

Optimizer: AdamW

Loss Function: Cross Entropy Loss

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- Macro F1-score



Results

Achieved stable accuracy across epochs

~74%

Macro F1-score indicates balanced performance across emotions (0.73)

Model generalizes well despite class imbalance

Emotion predictions successfully drive music recommendations

	precision	recall	f1-score	support
angry	0.70	0.73	0.71	3262
calm	0.55	0.70	0.61	2360
energetic	0.72	0.64	0.68	3615
happy	0.87	0.81	0.84	6630
sad	0.82	0.81	0.82	3829
accuracy			0.75	19696
macro avg	0.73	0.74	0.73	19696
weighted avg	0.76	0.75	0.76	19696

USER INTERFACE

- Built using **Streamlit**
- Allows users to input text describing their mood
- Displays detected emotion
- Shows recommended songs with Spotify preview links
- Provides a clean and interactive user experience

CHALLENGES & FUTURE SCOPE

CHALLENGES

1. CLASS IMBALANCE:

- Some emotions had significantly fewer samples

2. OVERFITTING:

- Validation loss increased after certain epochs
- Controlled using dropout and early stopping

3. MUSIC TAG NORMALIZATION:

- Large number of raw tags required simplification
- Manual mapping to emotion-relevant tags was necessary

FUTURE SCOPE

- Extend emotion detection to facial expressions
- Integrate Spotify API for dynamic playlists
- Improve recommendations using collaborative filtering
- Deploy as a full-stack web application
- Personalize recommendations using user history

CONCLUSION

- Successfully built an end-to-end emotion-based music recommendation system
- Combined deep learning with NLP and recommendation techniques
- Demonstrated real-time emotion detection and music suggestion
- Highlights the practical use of AI in enhancing user experience

THANK YOU